IYPT Presentation Checklist

What to Build, Slide by Slide, and Why

The Pinhole Project — 12-Minute Reporter Presentation

September 3, 2026

The Hard Constraints

- **12 minutes total** for the report presentation (per IYPT Regulations, Section VII) — fixed, not flexible.
- Immediately followed by Opponent questions — whatever is presented without its caveat visible becomes the easiest target.
- The Jury grades against a published rubric (attached image) — top marks explicitly require *both* a strong model *and* honest discussion of limitations/errors. Do not remove caveats to look more confident — it costs points, not gains them.

1 Slide 1 — Title / Mission (15–20 sec)

- Project name, one-line mission statement.
- One striking visual: a real generated array design image (not a stock photo) — signals rigor from the first second.

2 Slide 2 — The Problem (45 sec)

- Two devices exist separately: prescription glasses (correct focus) and sunglasses (attenuate light). State why combining them is genuinely hard, not just unexplored.
- One sentence on the stenopeic effect and why a naive pinhole array fails (ghosting, blind zones).

Rubric Link

Covers *"phenomenon explanation"* — top band requires *"shows physical insight,"* not just a restatement of the problem brief.

3 Slide 3–4 — The Physics: ASM vs. RSS (2–2.5 min)

- Defocus blur vs. diffraction blur, the classic trade-off.
- Most prior work (including the closest published paper) combines them by simple quadrature (RSS) — an approximation.
- Your model uses full Angular Spectrum wave-optics propagation — no shortcut, diffraction and defocus interact directly in one calculation.
- Show the blur-vs-aperture curve (you already have this figure).

Rubric Link

Covers *"theory/model"* — top band: *"detailed, complex, completely testable."* Naming the RSS shortcut explicitly, and why you avoided it, is what separates "good" from "detailed, demonstrative."

4 Slide 8–9 — Experiment (2 min)

- Show the ExP3 with-/without-pinhole comparison photo, and the radius-vs-displacement plot (clean, monotonic growth).
- State plainly: displacement in mm, not calibrated to diopters — this is a stated scope limit, not something to gloss over.
- If time allows: one sentence on the no-pinhole control comparison.

Rubric Link

Covers *"relevant experiments"* and *"errors analysis"* — the rubric explicitly names *"multitude of parameters examined"* and *"errors analysis"* as a top-band marker. Stating your calibration limit out loud is what earns this, not something that loses marks.

5 Slide 10 — Final Design + Limitations (1.5–2 min)

- The specific array design(s) your model recommends — real parameters, not placeholders.
- One slide, explicitly titled "What's Next" or similar: physical array fabrication, the two-pinhole ghost-margin test (not yet run), diopter calibration of the displacement method.

Rubric Link

Covers *"task fulfilment"* top band: *"greater extent than expected."* A visible, honest future-work list signals you know exactly where the project's edge is — which is different from, and scores higher than, implying it is finished.

6 Defence Prep — Separate From Slides, Prepare Cold

The rubric grades *"discussion with opponent"* and *"answers to jury"* independently from the report itself. Have tight, ready answers — not slide content, just talking points — for at least these:

1. **"How do you know your photographic measurements aren't just noise?"** — Answer with the saturation-flagging discipline, the connected-component method, and the honest scoping (illustrative, not precision-calibrated).
2. **"Why does your model disagree with clinical data?"** — State both candidate explanations from Slide 7, do not invent a resolution live.
3. **"Why trust your own ghost formula over the published one?"** — $R^2 = 0.995$ against direct wave-optics measurement, versus the published formula's own admitted saturation.
4. **"Have you built a physical array yet?"** — Answer honestly: single-aperture and displacement experiments done; full-array fabrication and the two-pinhole separation test are named, explicit future work.
5. **"Why RSS instead of full wave optics, unlike you?"** — Turn this around: you can name precisely *why* RSS is an approximation (assumes no interaction between geometric and diffractive blur) and what physical regime it breaks down in.

Rubric Link

Covers *"discussion with opponent"* and *"answers to jury, opponent, and reviewer's questions"* — top band across all three roles' rubrics rewards *"concise and correct,"* while *"inconclusive or too long"* or answers showing *"deep misconceptions"* actively lose points. A ready, rehearsed answer beats a better answer improvised badly.

Final Practical Checklist

- Total slide count: aim for 10–12 — roughly one per minute, leaves room for pacing.
- Every figure used should be one already generated and verified this project — do not create new illustrative graphics that lack a real number behind them.
- Rehearse against a clock at least twice — 12 minutes is tighter than it feels when planning.
- Bring the defence-prep answers as a printed card, not memorised from slides alone.